

James Hamilton

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EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

Manhattan, NY

Bachelor of Science (BS) in Computer Science | GPA: 3.68

December 2026

EXPERIENCE

Pace University (in partnership with NY State Hospitals)

New York, NY

NSF CISE REU Software Engineer – Applied Clinical AI

May 2026 - Present

- Engineered a native **Android (Java)** tablet application deployed to New York hospital networks, establishing a continuous feedback loop with medical professionals to refine clinical UI.
- Architected an on-device AI chatbot utilizing **local LLMs** and **RAG**, ensuring strict **HIPAA** compliance by preventing sensitive patient triage data from hitting third-party APIs.
- Developed an AI-powered Medication Scan feature utilizing embedded device cameras to rapidly capture, parse, and validate pharmacological data in live-triage environments.

Pace University, Seidenberg School of CSIS

New York, NY

AI Research Assistant (CRA Funded) - LLM Code Automation

April 2025 – September 2025

- Developed a **Python** benchmarking framework to evaluate **5 LLMs** across **50** structured software engineering tasks.
- Measured code correctness, runtime failures, and security violations using **Pytest**. Identified a **28 percent average** failure rate in authentication and input validation logic across models.
- Automated evaluation using unit tests and static analysis checks, reducing manual grading time by **60 percent** and improving result reproducibility.

Arcampass

Remote

Software Engineer (Founding Engineer)

February 2025 – January 2026

- Joined as a founding engineer at an early-stage ed-tech startup building an AI-driven personalized learning platform for higher education; built and shipped the initial platform using **Next.js (React)**, **Express**, and **MongoDB**.
- Designed **MongoDB** schemas for user accounts, sessions, and transactional data, then added compound indexes on the highest-traffic queries, cutting average database read latency from **320ms to 140ms**.
- Containerized the backend with **Docker**, deployed to **AWS EC2** behind **NGINX**, and layered in **Redis** caching, cutting deployment time from **12 minutes to 8**.
- Engineered an automated Python data extraction pipeline** to continuously monitor, parse, and ingest over **14,000** distinct NYU course records, extracting complex nested metadata including professor assignments and syllabus descriptions.

PROJECTS/INDEPENDENT HIRE

Master Parcel Data Engine - Next.js | Node.js | Go | PostgreSQL | AWS | Python | Redis | MongoDB

- Architected and deployed a real estate DaaS platform** featuring a hybrid **Go** and **Python (Pandas)** ETL pipeline that standardizes unstructured **MongoDB** market listings and **10+** municipal records (Ex: MapPLUTO) to the NYC BBL standard.
- Built a fullstack Next.js** frontend and backend to manage API routing, protect environment variables, and handle email-gated data streaming.
- Integrated a zero-cost architecture** using **Supabase (PostgreSQL)** for automated lead capture and CRM routing, alongside **Stripe** for seamless B2B monetization.

EthicsOS (Clinical AI Orchestration) - Python | FastAPI | PostgreSQL | OpenAI API | Supabase

- Architected an asynchronous clinical orchestration platform (**Top 10 Finalist, Cornell Hackathon**) using **Python (FastAPI)** and **OpenAI GPT-4o**, enforcing strict **JSON** schemas to automate medical insurance adjudication while eliminating AI hallucinations.
- Built a data enrichment ETL pipeline joining federal NPI registries with NYC Health metadata via heuristic matching, optimizing **PostgreSQL** with **JSONB** and GIN-indexing for rapid **FHIR** resource queries.
- Engineered a dual-output interoperability pipeline (**FHIR APIs** and e-Fax) and implemented a real-time observability layer via **Supabase** pub/sub to synchronize automated clinical decisions without blocking the main event loop.

Parkidsy (Seidenberg Creative Lab Program) - Next.js | Bun | Supabase (PostgreSQL) | TypeScript

- Architected a mobile-optimized **full-stack Next.js** and **Bun** web application wrapped inside **native iOS** and **Android WebViews** with a relational **Supabase (PostgreSQL)** schema for startup clients.

TECHNICAL SKILLS

- Languages:** Java, Python, TypeScript, SQL, Golang, JavaScript, HTML5, CSS
- Frameworks:** Spring boot, React.js, Next.js, Express.js, Node.js, FastAPI
- DevOps/Cloud:** Docker, AWS EC2, S3, Route 53, NGINX, Azure, CI/CD
- Databases:** MongoDB, PostgreSQL, Firebase, Redis, Supabase